



Electrical Machines-I

ECE-2107

Transformer-SL2

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“Guide us to the straight
path—”.

[Sura Fatihah]



Transformer Construction

Go through the article **32.2, 32.3, 32.4** of the book written by **B. L.**

Theraza to have a look at the transformer construction.



Transformer Phasor Diagram

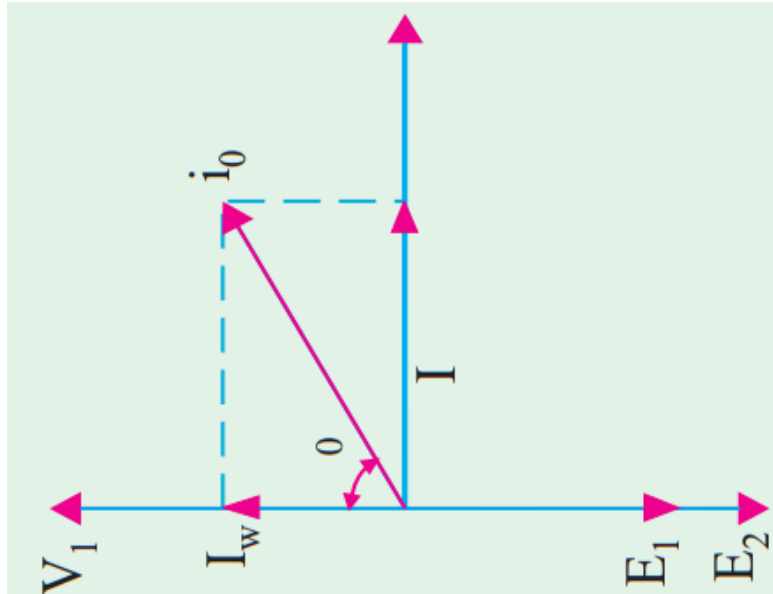
Two cases are consider

- When transformer is on no load
- When it is loaded

When an actual transformer is put on load, there is iron loss in the core and copper loss in the windings (both primary and secondary) and these losses are not entirely negligible. Even when the transformer is on no-load, the primary input current is not **wholly reactive**.

Transformer Phasor Diagram

No load condition



I_0 = Energizing current/ primary input current

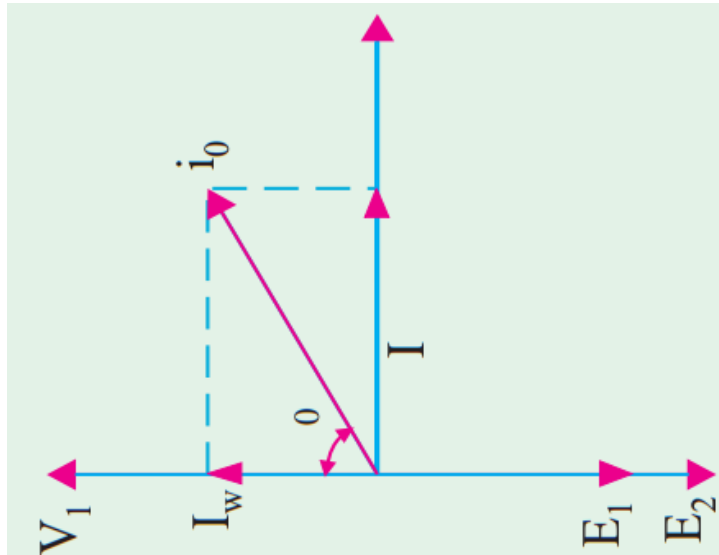
I_w = Iron loss current

I_μ = magnetizing current

θ_0 = No load power factor angle

Transformer Phasor Diagram

No load condition



No-load input power $W_0 = V_1 I_0 \cos \theta_0$

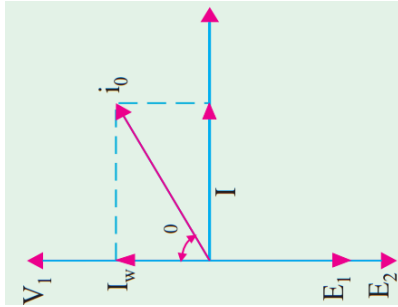
Iron loss current $I_w = I_0 \cos \theta_0$

magnetizing current $I_\mu = I_0 \sin \theta_0$

For preparing your answer you can go through the article **14.5** of the book written by “**Rosenblatt**”

Transformer Phasor Diagram

No load condition

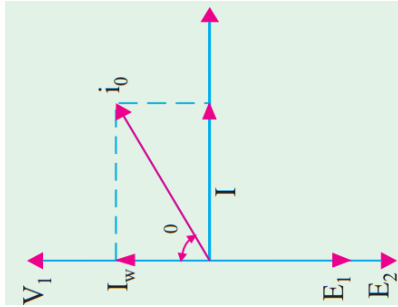


The followings should be noted:

- ❑ The no-load primary current I_0 is very small as compared to the full-load primary current. It is about **1 %** of the **full-load** current.
- ❑ As the permeability of the core varies with the instantaneous value of the exciting current, the wave of the exciting or magnetizing **current** is not **truly sinusoidal** and should not be represented by a **vector**.

Transformer Phasor Diagram

No load condition



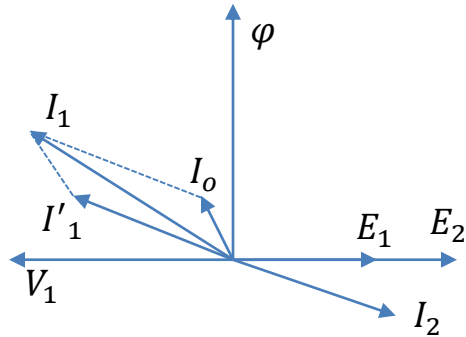
- ❑ As I_0 is very **small**, the no-load primary **Cu loss** is negligibly **small** which means that no-load primary input is practically equal to the iron loss in the transformer.
- ❑ As it is principally the **core-loss** which is responsible for **shift** in the current vector, angle θ_0 is known as hysteresis angle of advance.

Problems

Practice example **14.4** of **Rosenblatt** and
32.9 of B. L. **Theraza** and also the related
tutorial problem.

Transformer Phasor Diagram

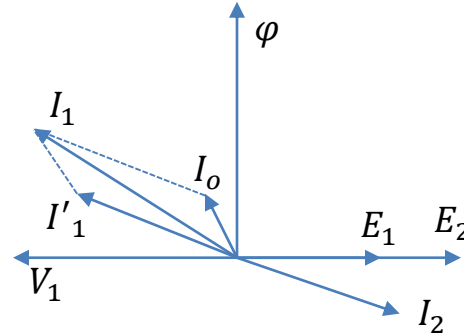
Loaded condition



- When **load** is applied to the **secondary** terminals, in accordance with the Lenz's law, the current that flows through the windings must act in such a direction as to **oppose** the flux set up by the **primary current**.
- When the flux is momentarily **reduced**, the induced **emf** in the **primary** winding is also **reduced** and as a result **more current** flows in the **primary** winding.
- The **increased** primary **current** will cause the flux to increase to its **original** value. When more load current flows in the secondary, the process is repeated and the primary current will again increase.

Transformer Phasor Diagram

Loaded condition



- ❑ The load current I_2 is shown lagging the secondary induced voltage E_2 . I'_1 is the current that flows in the primary winding to balance the demagnetizing effect of I_2 .
- ❑ Since flux ϕ remains constant, I_0 must be the same current that energizes the transformer at no-load.
- ❑ I_1 , the actual current that flows in the primary, is therefore the phasor sum of currents I'_1 and I_0



Transformer Phasor Diagram

Loaded condition

When the load is purely resistive

For preparing your answer you can go through the article 32.10 of the book written by “B. L. Theraza”

Problems

Practice example **32.12, 32.13** and **32.14**
of B. L. **Theraza** and also the related tutorial
problem.